

Skills checks cover pre-requisite material needed for the third unit of the course, applications of derivatives. Answer the following questions neatly and completely. Calculators are not allowed.

1. Given $3(x^2 + 4)^{1/3} - 2x(x^2 + 4)^{-2/3}$, factor out $(x^2 + 4)^{-2/3}$ and write your final answer using only positive exponents.

$$3(x^2 + 4)^{1/3} - 2x(x^2 + 4)^{-2/3} \Leftrightarrow (x^2 + 4)^{-2/3} [3(x^2 + 4) - 2x] \Leftrightarrow \frac{3(x^2 + 4) - 2x}{(x^2 + 4)^{2/3}}$$

2. Given $\sqrt{x+1} - 5(x+1)^{-1/2}$, factor out $(x+1)^{-1/2}$ and write your final answer using only positive exponents.

$$\sqrt{x+1} - 5(x+1)^{-1/2} \Leftrightarrow (x+1)^{-1/2} [(x+1) - 5] \Leftrightarrow \frac{(x+1) - 5}{(x+1)^{1/2}}$$

3. Given $7(2x-3)^{2/3} - x(2x-3)^{-1/3}$, factor out $(2x-3)^{-1/3}$ and write your final answer using only positive exponents.

$$7(2x-3)^{2/3} - x(2x-3)^{-1/3} \Leftrightarrow (2x-3)^{-1/3} [7(2x-3) - x] \Leftrightarrow \frac{7(2x-3) - x}{(2x-3)^{1/3}}$$

4. Solve the inequality $\frac{2x+1}{x^2} \geq 0$. Give your answer in interval notation.

We know that $\frac{2x+1}{x^2} = 0$ when $2x+1=0 \Leftrightarrow x = -\frac{1}{2}$

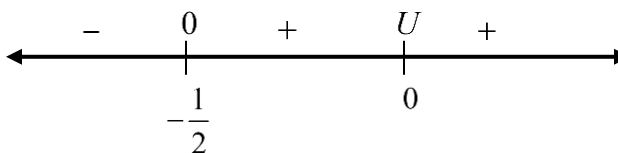
We know that $\frac{2x+1}{x^2}$ is undefined when $x^2 = 0 \Leftrightarrow x = 0$

Therefore, we put these values on a number line (representing the x -axis) and we check to see the behavior of the expression $\frac{2x+1}{x^2}$ on the intervals on either side (and in between) these values.

If $x = -1$ then $\frac{2x+1}{x^2} < 0$

If $x = -\frac{1}{4}$ then $\frac{2x+1}{x^2} > 0$

If $x = 1$ then $\frac{2x+1}{x^2} > 0$



Since we are interested in when the expression is greater than (or equal to) zero we have the solution set:

$$\left[-\frac{1}{2}, 0\right) \cup (0, \infty)$$

5. Solve the inequality $(x-2)(x+1)^3 > 0$. Give your answer in interval notation.

We know that $(x-2)(x+1)^3 = 0$ when $x = 2$ or $x = -1$

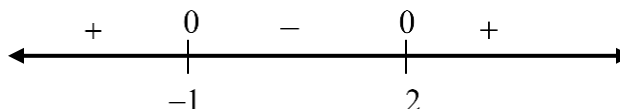
We know that $(x-2)(x+1)^3$ is never undefined.

Therefore, we put these values on a number line (representing the x -axis) and we check to see the behavior of the expression $(x-2)(x+1)^3$ on the intervals on either side (and in between) these values.

If $x = -2$ then $(x-2)(x+1)^3 > 0$

If $x = 0$ then $(x-2)(x+1)^3 < 0$

If $x = 5$ then $(x-2)(x+1)^3 > 0$



Since we are interested in when the expression is greater than zero we have the solution set:

$$(-\infty, -1) \cup (2, \infty)$$

6. Solve the inequality $e^x(x-2) \leq 0$. Give your answer in interval notation.

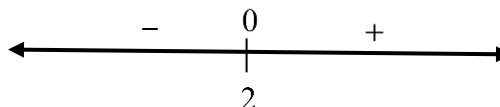
We know that $e^x(x-2) = 0$ when $x = 2$

We know that $e^x(x-2)$ is never undefined.

Therefore, we put these values on a number line (representing the x -axis) and we check to see the behavior of the expression $e^x(x-2)$ on the intervals on either side (and in between) these values.

If $x = 0$ then $e^x(x-2) < 0$

If $x = 5$ then $e^x(x-2) > 0$



Since we are interested in when the expression is less than (or equal to) zero we have the solution set:

$$(-\infty, 2]$$

7. Solve the equation: $x \ln(2) - x^2 = 0$

$$x \ln(2) - x^2 = 0 \rightarrow x[\ln(2) - x] = 0 \rightarrow x = 0 \text{ or } x = \ln(2)$$

8. Solve the equation: $e^x - xe^x = 0$

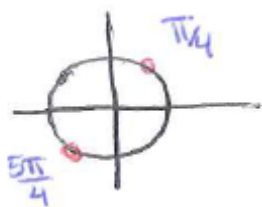
$$e^x - xe^x = 0 \rightarrow e^x[1 - x] = 0 \rightarrow x = 1$$

9. Solve the equation: $x^2 \ln(x) - 4 \ln(x) = 0$.

$$x^2 \ln(x) - 4 \ln(x) = 0 \rightarrow \ln(x)[x^2 - 4] = 0 \rightarrow \ln(x) = 0 \text{ or } x^2 - 4 = 0 \rightarrow x = e \text{ or } x = \pm 2$$

10. Solve the equation $6 \tan(\theta) - 6 = 0$ for θ .

We start by isolating $\tan(\theta)$. We see that $6 \tan(\theta) = 6 \rightarrow \tan(\theta) = 1$



$\theta = \frac{\pi}{4}$ & $\theta = \frac{5\pi}{4}$ are both possible solutions.

Any angle coterminal with these angles are also solutions.

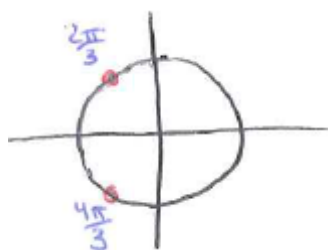
Therefore our solutions are: $\theta = \frac{\pi}{4} + 2\pi k$ where k is an integer.
 $\theta = \frac{5\pi}{4} + 2\pi k$

Since the locations of these solutions are separated by a distance of π we can combine these equations into a single equation.

$$\theta = \frac{\pi}{4} + \pi k \text{ where } k \text{ is an integer.}$$

11. Solve the equation $\frac{1}{2} + \cos(\theta) = 0$ for θ .

We would like to start by first isolating the $\cos(\theta)$. We see that $\cos(\theta) = -\frac{1}{2}$.



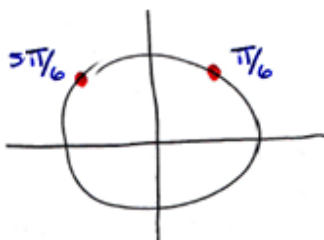
$\theta = \frac{2\pi}{3}$ & $\theta = \frac{4\pi}{3}$ are both possible solutions.

Any angle coterminal with these angles are also solutions.

Therefore our solutions are: $\theta = \frac{2\pi}{3} + 2\pi k$ where k is an integer.
 $\theta = \frac{4\pi}{3} + 2\pi k$

12. Solve the equation $2 \sin(\theta) - 1 = 0$ for θ .

We would like to start by first isolating the $\sin(\theta)$. We see that $\sin(\theta) = \frac{1}{2}$.



$\theta = \frac{\pi}{6}$ & $\theta = \frac{5\pi}{6}$ are both possible solutions.

Any angle coterminal with these angles are also solutions.

Therefore our solutions are: $\theta = \frac{\pi}{6} + 2\pi k$ where k is an integer.
 $\theta = \frac{5\pi}{6} + 2\pi k$

13. Find the volume of a cylinder with height of 5cm and a diameter of 2cm .

The volume of a cylinder is given by $V = \pi r^2 h$.

Therefore, $V = \pi (1)^2 5 = 5\pi$

14. Find the radius of a circle that has an area of 12in^2 .

The area of a circle is given by $A = \pi r^2$.

Therefore, $12 = \pi r^2 \rightarrow r = \sqrt{\frac{12}{\pi}}$

15. Determine the length of the remaining side of the triangle at the right.

The Pythagorean Theorem states that in a right triangle $a^2 + b^2 = c^2$, where c represents the hypotenuse of the triangle.

Therefore, $a^2 + (6)^2 = (8)^2 \rightarrow a^2 + 36 = 64 \rightarrow a^2 = 28 \rightarrow a = \sqrt{28}$

